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Module: Programming 2B

Section B: API Endpoint Plan (Updated)

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new account. The role selected decides the id prefix (O or P).	Guest	{ fullName, email, password, role, dateOfBirth, gender } (dateOfBirth and gender only for Participants)	201 Created – user profile with generated id and role (no password) / 400 Bad Request – validation errors / 409 Conflict – email already registered
POST	/api/auth/login	Lets a **registered** Organiser or Participant log in (or log back in after logging out). The password is hashed and compared with the stored hash.	Guest (must already have an account)	{ email, password }	200 OK – session started, returns id, name, role / 401 Unauthorized – invalid credentials or no account found (register first)

		Unregistered guests are refused and must register first.			
POST	/api/auth/logout	Ends the current session. The user returns to guest access.	Any User	None	200 OK – session ended / 401 Unauthorized

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves the logged-in user's own profile.	Any User	None	200 OK – profile object / 401 Unauthorized
PUT	/api/users/profile	Updates the logged-in user's own profile.	Any User	{ fullName, email } (Participants may also send dateOfBirth, gender)	200 OK – updated profile / 400 Bad Request / 401 Unauthorized / 409 Conflict – email in use

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all events (guests can browse).	Guest	None	200 OK – list of events

GET	/api/events/{id}	Retrieves a single event's details.	Guest	None	200 OK – event object / 404 Not Found
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, eventType }	201 Created – event object / 400 Bad Request / 401 Unauthorized / 403 Forbidden
PUT	/api/events/{id}	Updates an event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, eventType }	200 OK – updated event / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

4. Categories

HTTP Method	**Route**	**Description**	**Role Required**	**Request Body**	**Expected Response**
GET	/api/events/{eventId}/categories	Lists all categories for an event.	Guest	None	200 OK – list of categories / 404 Not Found

POST	/api/events/{eventId}/categories	Creates a category under an event owned by the logged-in Organiser.	Organiser	{ categoryName, distance }	201 Created – category object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
PUT	/api/categories/{id}	Updates a category belonging to the Organiser's event.	Organiser	{ categoryName, distance }	200 OK – updated category / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/categories/{id}	Deletes a category belonging to the Organiser's event.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

5. Event Enrolments

HTTP Method	**Route**	**Description**	**Role Required**	**Request Body**	**Expected Response**

POST	/api/enrolments	Enrols the logged-in Participant into a category. Guests must register and log in first.	Participant	{ categoryId }	201 Created – enrolment object / 400 Bad Request / 401 Unauthorized – not logged in / 403 Forbidden – not a Participant / 404 Not Found / 409 Conflict – already enrolled
GET	/api/enrolments/my	Lists all enrolments belonging to the logged-in Participant.	Participant	None	200 OK – list of enrolments / 401 Unauthorized / 403 Forbidden
PUT	/api/enrolments/{id}/cancel	Cancels the Participant's own enrolment (status set to Cancelled).	Participant	None	200 OK – updated enrolment / 401 Unauthorized / 403 Forbidden / 404 Not Found
GET	/api/events/{eventId}/enrolments	Lists all enrolment	Organiser	None	200 OK – list of

		s for an event owned by the logged-in Organiser.			enrolments / 401 Unauthorized / 403 Forbidden / 404 Not Found

6. Results

HTTP Method	**Route**	**Description**	**Role Required**	**Request Body**	**Expected Response**
POST	/api/results	Captures a finish time and position for a Participant's enrolment in the Organiser's event.	Organiser	{ enrolmentId, finishTime, position }	201 Created – result object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – result already captured
GET	/api/results	Lists the results of **all participants** across every event owned by the	Organiser	None	200 OK – list of results (participant, event, category, finish time, position) / 401

		logged-in Organiser, in one list.			Unauthorized / 403 Forbidden
GET	/api/events/{eventId}/results	Lists the results of all participants for one event owned by the logged-in Organiser.	Organiser	None	200 OK – list of results / 401 Unauthorized / 403 Forbidden / 404 Not Found
GET	/api/results/my	Lists all results belonging to the logged-in Participant.	Participant	None	200 OK – list of results / 401 Unauthorized / 403 Forbidden
GET	/api/results/{id}	Retrieves a single result (the owning Participant or the Organiser who captured it).	Any User	None	200 OK – result object / 401 Unauthorized / 403 Forbidden / 404 Not Found

7. Routes (a category can have zero or many routes)

HTTP Method	**Route**	**Description**	**Role Required**	**Request Body**	**Expected Response**
GET	/api/categories/{categoryId}/routes	Lists all route	Guest	None	200 OK – list of

		maps for a category.			routes / 404 Not Found
POST	/api/categories/{categoryId}/routes	Adds a route map to a category owned by the logged-in Organiser.	Organiser	{ mapFileUrl }	201 Created – route object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
PUT	/api/routes/{id}	Updates a specific route map.	Organiser	{ mapFileUrl }	200 OK – updated route / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/routes/{id}	Deletes a specific route map.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

Access Summary

Area	Guest	Participant	Organiser
Register / Login	Yes	–	–
View events, categories, routes	Yes	Yes	Yes
View / update own profile	No	Yes	Yes
Enrol in a category, view own enrolments, cancel	No	Yes	No
View own results	No	Yes	No
Create / update / delete events, categories, routes	No	No	Yes
View enrolments for own events	No	No	Yes
Capture results	No	No	Yes
View results of all participants (own events)	No	No	Yes